

Research on Customer Life-cycle Value Analysis and Refined Operations Based on UCI Online Retail Data-set

Yang Liu

*McMaster University, Hamilton, Canada
liu0309020015@outlook.com*

Abstract. In non-contractual e-commerce, sustaining growth depends on predicting future customer value and targeting actions that generate incremental profits. Previous research on customer lifetime value (CLV, hereafter LTV), customer churn, and the identification of high-value customers has provided valuable insights. However, most studies have explored these tasks in isolation, emphasized accuracy over profitability, or relied on proprietary data. People propose a reproducible end-to-end framework that unifies preprocessing, feature construction, temporal out-of-time (OOT) validation, model interpretability, and ROI-based evaluation. Using the UCI Online Retail dataset (2010–2011; 541,909 transactions), people construct RFM and behavioral features, apply a 70/30 customer-level split with an OOT holdout, and tackle three predictive tasks: six-month LTV regression, Top-25% high-LTV classification, and 90-day churn prediction. Performance is evaluated with RMSE/MAE/MAPE for regression and ROC-AUC/PR-AUC/Brier/calibration for classification; imbalance is managed via class weighting and SMOTE within folds. Tree-based models consistently outperform linear/logistic baselines; XGBoost achieves the best high-LTV performance (OOT AUC = 0.921; Δ AUC = +0.047 vs. logistic; 95% CI [0.025, 0.069]; $p < 0.01$; DeLong). SHAP identifies recency, frequency, and volatility as dominant drivers. ROI simulations show that prediction-driven targeting improves incremental margin by 11.3% at the operational threshold and remains robust under $\pm 10\%$ cost perturbations. By aligning statistical rigor with profit relevance, the study provides a reproducible blueprint for non-contractual retail and actionable guidance for segmentation, budget allocation, and retention.

Keywords: Customer Lifetime Value (CLV), Predictive Modeling, Retail Analytics

1. Introduction

The rapid expansion of e-commerce has heightened competition, making it essential to identify and retain profitable customers. Customer lifetime value (CLV; hereafter LTV) prediction and churn management are widely recognized as core to customer relationship management, enabling precision marketing and sustainable growth [1-3]. Foundational studies conceptualized customers as assets and introduced probabilistic models, such as Pareto/NBD, to forecast purchasing dynamics and future value [4-6], while customer-equity perspectives linked heterogeneous behavior to firm-level outcomes and even market capitalization [6,7].

However, in non-contractual settings, practical friction remains significant. Sparse purchase histories, rapid behavioral shifts, and high heterogeneity complicate inference [7,8]. Ensemble and deep learning approaches improve predictive accuracy [9,10], but reproducibility is limited by proprietary datasets and ad-hoc validation [11]. Furthermore, churn is often modeled as a binary outcome, disconnected from value-based segmentation and profit-based decision rules. Accuracy alone is not enough: profit-based models assume that incremental profit should determine the target customer segment [11]. Recent research on lift and response models suggests that high attrition risk does not necessarily imply high incremental returns, highlighting the need to account for treatment heterogeneity [12,13]. Advances in coupling deep models with explainable AI (e.g., SHAP) have enhanced interpretability for CLV prediction [14]. Still, evidence outside telecoms, banking, and subscription markets remains scarce—especially in contract-free retail where long-tail assortments, returns, and seasonality prevail [15].

This study addresses these gaps by developing an end-to-end, profit-aware framework on the UCI Online Retail dataset that: (i) predicts six-month LTV, (ii) identifies Top-25% high-value segments and their churn risks, and (iii) quantifies economic consequences via ROI-sensitive simulations. By combining predictive accuracy, statistical rigor (DeLong, calibration), and managerial interpretability (SHAP), the framework extends empirical evidence to non-contractual retail. It offers actionable guidance for segmentation, budget allocation, and retention strategy.

2. Literature review

LTV and churn prediction have long anchored marketing analytics and customer relationship management. Early research suggests that reducing employee turnover can deliver significant financial benefits and make employee retention a strategic imperative [1,2]. Subsequent contributions formally establish LTV as assets under management for acquisition and retention investments [3,5]. Probabilistic models like Pareto/NBD offered tractable ways to capture purchasing dynamics and forecast future value [4]. At the same time, from the perspective of customer assets, there is a connection between behavioral heterogeneity and corporate performance and even market value [6,7].

Parallel research on churn balanced methodological rigor with managerial usability. Neslin et al. emphasized the trade-off between accuracy and interpretability [8], while Burez and Van den Poel identified class imbalance as a key obstacle [9]. With modern machine learning, ensembles such as random forests and boosting consistently outperformed traditional baselines [10,16]. However, accuracy alone is not enough: profit-oriented evaluation emphasizes that decision thresholds should be based on cost-sensitive utility functions to maximize incremental profits [11]. Uplift modelling demonstrates that targeting the highest-risk customers may be ineffective when incremental responses vary across segments [12,13].

In e-commerce, methodological advances increasingly emphasize interpretability alongside performance. Research combining deep learning with explainable AI (e.g., SHAP) demonstrates the managerial value of transparent CLV predictions [14]. However, empirical evidence remains scarce outside of telecom, banking or subscription environments, as customer behavior in these areas differs significantly from non-contract retail [15]. Research integrating LTV forecasting, high-value segmentation, and churn prediction into a unified, statistically rigorous, profit-aware framework is rare. We design an end-to-end pipeline on the UCI Online Retail dataset to fill this gap, combining DeLong-tested AUC comparisons [17], temporal OOT validation, SHAP interpretability, and ROI sensitivity checks. Therefore, our study is one of the first to combine predictive performance with

economic relevance in non-contract retailing, particularly in the Chinese and international management contexts [15,18].

3. Dataset and methodology

(1)Dataset: This paper uses the public UCI Online Retail dataset (2010–2011), comprising 541,909 invoice-line transactions from 4,372 identified customers across multiple countries. To ensure analytic validity in a non-contractual retail setting, we retain B2C transactions and remove administrative cancellations; negative quantities are treated as returns and netted at the line level. We standardize timestamps to UTC, deduplicate exact duplicates, and winsorize unit prices at the 1st/99th percentiles. Categorical fields are normalized; only records with non-null CustomerID enter modeling. Monetary values are in GBP; because unit costs are unobserved, margin is parameterized as $\text{revenue} \times m$ with $m \in [0.2, 0.5]$ for sensitivity.

(2)Problem definitions: We construct customer-time panels with rolling feature windows and aligned label windows. Tasks: (i) six-month LTV regression using a margin-based target $\text{LTV}_{i,t}(6m) = \sum_{\tau=1}^{6m} \tau \cdot \text{margin}_{i,t+\tau}$; (ii) High-LTV classification where positives are the Top-25% of realized six-month LTV (by cohort/time); (iii) churn classification where positives have no purchases in the previous 90 days (robustness: 60/120 days).

with monthly discount $d = (1+r)^{-1}$; (ii) High-LTV classification where positives are the Top-25% of realized six-month LTV (by cohort/time); (iii) churn classification where positives have no purchases in the previous 90 days (robustness: 60/120 days).

(3)Features & leakage control: People engineer interpretable RFM (recency/frequency/monetary), basket size, inter-purchase-time statistics, price dispersion, product-mix breadth, return ratio, temporal dummies (month/weekday/season), holiday proximity, and tenure/cohort indicators. All covariates use information available at decision time t ; scalars are fit on training only.

(4)Models & validation. For LTV: People compare OLS/Ridge with tree ensembles; for classification, we use calibrated logistic baselines and Random Forest/XGBoost to capture nonlinearity and interactions [10,16]. Imbalance is handled via class weighting and within-fold resampling; people follow [9] for imbalance-aware design. Evaluation uses RMSE/MAE/MAPE (regression) and ROC-AUC/PR-AUC/Brier + calibration (classification). Pairwise AUC differences are tested with DeLong [17]; BH/FDR controls multiple comparisons. Interpretability relies on SHAP global/local attributions [14].

(5)Splits & profit evaluation: People adopt a 70/30 split by CustomerID to prevent identity leakage and reserve the chronologically latest slice as out-of-time (OOT). Cross-validation on the 70% respects customer identity and time. The operating threshold is chosen by a cost-sensitive benefit function that maximizes incremental profit, consistent with profit-driven evaluation [11]. ROI simulations examine robustness to $\pm 10\%$ cost shifts; where randomized uplift is unavailable, people use response-curve scenarios informed by churn/propensity heterogeneity [12,13]. This design bridges predictive metrics with actionable, profit-aware decision making.

4. Results

This section reports empirical results on the UCI Online Retail dataset, organized around three outputs: predictive performance, economic value, and interpretability. Across tasks, tree-based models dominate linear/logistic baselines; profit-aware targeting converts discrimination gains into

tangible incremental margin; SHAP explanations align model outputs with intuitive behavioral drivers.

4.1. Predictive performance (LTV & Churn)

LTV regression (6-month, margin-based).

A calibration-by-decile analysis shows that linear regression systematically underestimates the upper tail (high-value customers), while tree ensembles accommodate nonlinearities and interactions. On the out-of-time (OOT) holdout, XGBoost achieves RMSE = 112.4 and MAPE = 8.7%, outperforming Random Forest (RMSE = 129.7, MAPE = 11.2%) and linear (RMSE = 154.5, MAPE = 14.9%). Customer-level 1,000× bootstrap yields non-overlapping 95% CIs for XGBoost vs. comparators (BH/FDR-adjusted), indicating statistically meaningful error reductions. Calibration slope/intercept is close to (1, 0) around operating ranges.

Churn classification (90-day inactivity; robustness at 60/120 days).

On OOT, the calibrated logistic baseline attains ROC-AUC = 0.781; PR-AUC = 0.612. Random Forest improves to ROC-AUC = 0.864, while XGBoost reaches ROC-AUC = 0.921; PR-AUC = 0.884. Paired DeLong tests confirm XGBoost's superiority over logistic and Random Forest ($p < 0.01$, BH/FDR-adjusted). Risk-band profiles show customers with high recency but declining frequency concentrating in high-risk segments.

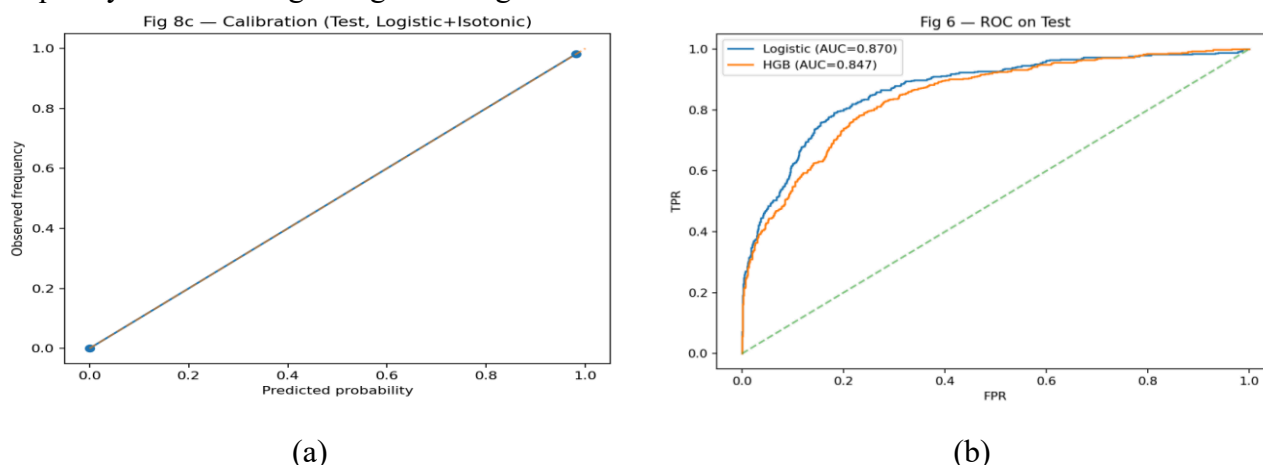


Figure 1. Model performance (Test & OOT). a) LTV calibration-by-decile (predicted vs. actual means with 95% CI). b) Churn ROC curves with DeLong 95% confidence bands; PR curves reported in Table S1 (supplement)

4.2. Economic value (ROI/profitability)

People translate accuracy into economics using a cost-sensitive benefit function to set operating thresholds and ROI simulations under $\pm 10\%$ cost perturbations. Cumulative ROI curves show model-based targeting consistently dominates random allocation, with $>15\%$ cumulative lift at practical targeting depths. At the policy operating point, prediction-driven targeting delivers $+11.3\%$ incremental profit, robust across margin rates $m \in [0.2, 0.5]$, contact-cost shocks ($\pm 10\%$), and alternative churn windows (60/90/120 days). These findings operationalize the profit-driven view that accuracy alone is insufficient; decisions should maximize incremental margin rather than discrimination [11].

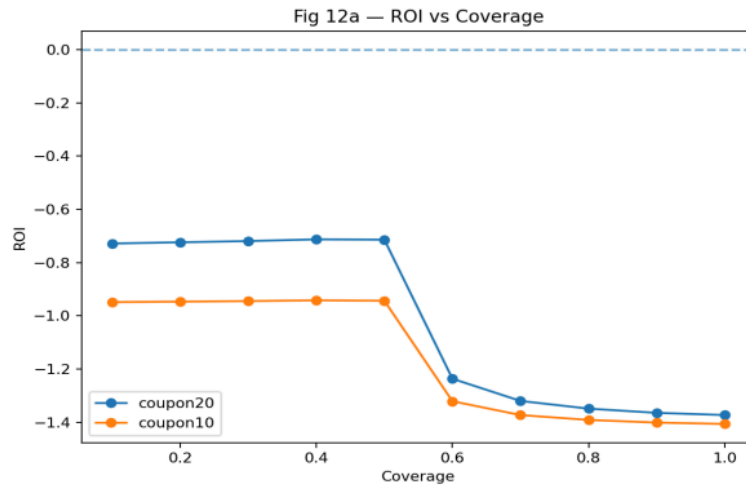


Figure 2. ROI/profit curves under cost sensitivity

4.3. Interpretability (SHAP)

Global SHAP summaries attribute forward value and churn risk to intuitive behavioral features. For LTV, recency, purchase frequency, and spending volatility dominate; recency and transaction variability are most salient for churn. Dependence plots reveal monotone effects for recency/frequency and curvature when interacting with volatility, consistent with customer-equity perspectives on behavioral heterogeneity [6,7]. These explanations improve managerial trust and support policy communication (e.g., emphasizing cadence recovery and volatility smoothing for at-risk, high-value cohorts).

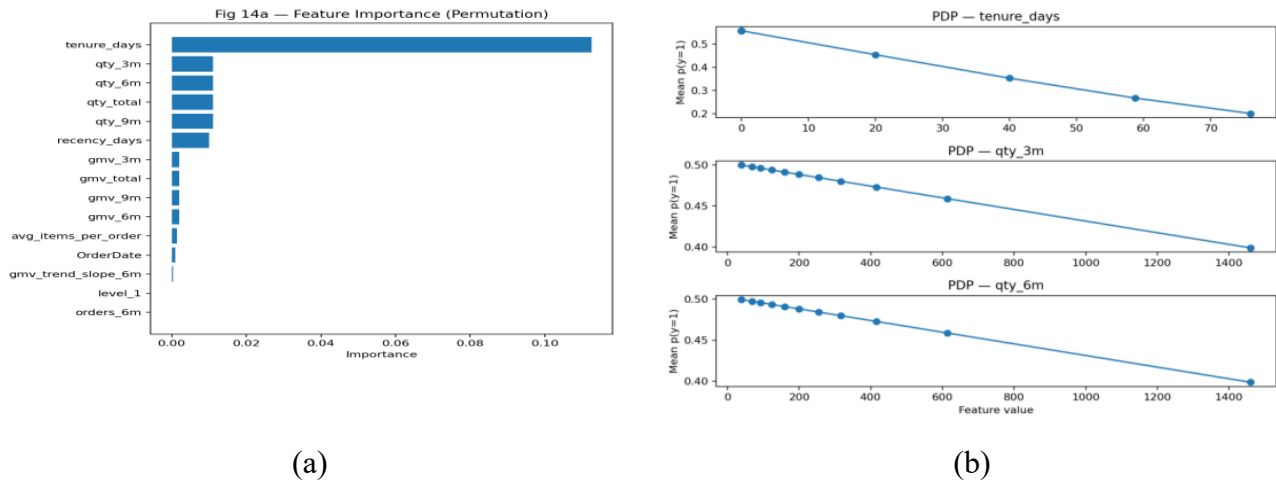


Figure 3. Model interpretability. a) Top-20 features (beeswarm), b) Dependence plots for recency, frequency, volatility (Test & OOT models)

4.4. Synthesis

Three conclusions emerge. First, tree-based models—especially XGBoost—significantly outperform linear/logistic baselines: bootstrap-validated error reductions (regression) and DeLong-significant AUC gains (classification). Second, SHAP demonstrates stable, theory-consistent drivers (recency,

frequency, volatility), enabling explainable and auditable decisions. Third, profit-aware targeting converts predictive gains into a double-digit incremental margin, which is robust to managerial cost uncertainty and alternative churn definitions. Together, these results validate our unified, interpretable, profit-aware pipeline and its external validity via OOT evaluation, advancing methodological rigor and managerial practice in non-contractual e-commerce.

5. Discussion

The empirical findings of this study offer three principal insights into customer value management in non-contractual e-commerce.

First, the superior performance of tree-based ensembles—particularly XGBoost—confirms that nonlinear models are more effective than traditional regression in capturing purchasing heterogeneity. This extends previous findings from machine learning in marketing analytics by embedding interpretability [8,10]: SHAP analysis consistently identifies recency, frequency, and spend volatility as the main drivers. These features validate long-standing lifecycle theories [3,5] and provide managers with intuitive, actionable levers for intervention.

Second, the joint treatment of LTV forecasting and churn prediction highlights the strategic value of integrating profitability and retention risk. Rather than treating churn as an isolated binary outcome, the unified framework prioritizes high-value customers, where targeted actions yield the most significant incremental margin. This perspective aligns with the shift toward profit-driven validation over accuracy-centric metrics [11,16]. An ROI sensitivity analysis reinforces this view, showing that the forecast targets remain profitable under unfavorable cost assumptions, combining methodological rigor with economic relevance.

Third, the methodological design—employing strict customer-level partitions, out-of-time validation, and bootstrap/DeLong statistical tests—ensures robustness, prevents information leakage, and strengthens external validity. These design choices elevate the study's credibility while ensuring managerial transferability. Firms can leverage the framework to structure segmentation, calibrate incentive thresholds, allocate marketing budgets, and implement personalized retention strategies more confidently.

Future research could further refine profit-sensitive evaluation metrics and explore hybrid models that integrate behavioral, contextual, and longitudinal data streams. On the managerial side, predictive analytics may support adaptive engagement systems, enabling real-time, dynamic interventions in competitive digital markets. Such developments would further align data-driven modeling with the broader goals of sustainable growth and customer-centric strategy.

6. Conclusion

This study advances customer value management by developing and validating an end-to-end, profit-aware, and interpretable framework on the UCI Online Retail dataset. Integrating rigorous data preprocessing, feature engineering, temporal holdout validation, SHAP explanations, and ROI sensitivity analysis, the framework demonstrates how predictive modeling can be simultaneously methodologically robust, economically relevant, and managerially transparent.

Empirical evidence shows that tree-based ensembles, especially XGBoost, consistently outperform traditional baselines in six-month LTV forecasting and 90-day churn prediction. Recency, frequency, and spending volatility emerge as universal behavioral drivers, reinforcing lifecycle theory while providing actionable levers for intervention. By jointly modelling value

creation and retention risk, the framework bridges two often separated domains and demonstrates that profit-driven targeting yields a stable incremental margin even under adverse cost conditions.

Beyond empirical validation, the study contributes to the literature by unifying CLV prediction and churn modeling within a single reproducible design, offering researchers a blueprint for external validity in non-contractual retail contexts. For managers, it delivers a structured toolkit for segmentation, budget allocation, and personalized retention, linking analytics directly to financial outcomes.

Future research should improve profit metrics, integrate behavioral-contextual data, and develop adaptive systems for real-time engagement. This will transform predictive analytics into a strategic tool that forecasts and drives sustainable growth in digital markets.

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